The Hashemite University

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Object Oriented Programming 1

Lab Manual

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Worksheet One: Working with Java

Question#1 (Output)

Write a program to print the following message "Welcome to Java class".

Question#2 (Swap)

Consider the following two variables:

int x=10;

int y=50;

Write a program to print the values of x and y, then swap the two values using a third variable, then, reprint the new values of x and y.

Question #3 (Precedence Rules)

Write a program to calculate and print the result of the equation below, consider the following variables:

int x=1; int y=1;

int a=1; int b=1; int c =1;

$$\frac{3+4x}{5}-\frac{10(y-5)(a+b+c)}{x}+9(\frac{4}{x}+\frac{9+x}{y})$$

Question #4 (Digits)

Assume that

int x= 123;

Write a program that calculates and prints the sum of all digits in a 3-digits number. Assume int x =123 is our number.

For example, x=123, the sum of all its digits is 1+2+3 =6.

(Hint): Use the % operator to extract digits, and use the / operator to remove the extracted digit. For instance, 123 % 10 = 3 and 123 / 10 = 12

Worksheet Two: Introduction to Java (Input/Output)

Question#1 (Using Console)

Write a program that reads the marks of three students from the console, then calculates and displays their average using the console too.
Hint: use the following piece of code:

```
Scanner s = new Scanner (System.in);
int h=s.nextInt();
```

Do not forget to import java.util.Scanner;

Question #2 (Reading characters):

Write a program that reads any character using the console then prints the corresponding Ascii code. For example, if the user enters 'a', the output will be 97.

Rewrite the program using Dialog Boxes.

Question #3 (Using Dialog Boxes)

Write a program that reads a Fahrenheit degree in double from an input dialog box, then converts it to Celsius and displays the result in a message dialog box.
Hint: celsius = (5/9) * (fahrenheit - 32), use the following piece of code:

```
String s = JOptionPane.showInputDialog(null,"Enter the Fahrenheit
degree");
double fahrenheit =Double.parseDouble(s);
```

Do not forget to import `javax.swing.JOptionPane ;`

Rewrite the program using console.

Question #4 (Math Class):

Write a program that reads the radius of a circle using dialog boxes then calculates and displays its area.
Hint : cir _area =PI*radius^2, use Math.PI constant and Math.pow(double,double) method.

Rewrite the program using console.

Worksheet Three: Selection Statements

Question #1 (if/else)

Write a program that reads a student mark in double and prints **pass** if mark is greater than or equal 50 otherwise prints **fail**.

Question #2 (nested If/else)

Write a program that reads an integer number and prints if it is an odd or even number, positive or negative number.

Question #3 (Conjunction Operators)

Write a program that reads an integer number and determines whether it is divisible by both 5 and 6, whether it is divisible by either 5 or 6, and whether it is divisible by either 5 or 6 but not both. Hint: Use the (&&, || and ^ operators).

Question #4 (Salary):

Write a program that reads the basic salary in double (using dialog boxes). The program should calculate the net salary according to the formula:

Net_salary= basic_salary – basic_salary*tax

- Tax equals 0.18 if the basic salary is greater than 1000.
- Tax equals 0.12 if the basic between 500 and 1000.
- Otherwise tax equals 0.08

The net salary should be displayed in message dialog box with no icon and the title should be "Net **Salary**".

Question #5 (Switch)

Write a program that reads two integers and a mathematical operation character (+,-,*,/) then find the proper result (Hint: use switch statement and dialogboxes). For instance, if user entered 3, 2 and '+' the result should be (5).

Question #6 (Rewrite if into switch)

Rewrite the following if statement into switch statement. Initialize x to 0 and a to 3, and display the value of x after executing the switch statement.

```
if (a == 1)
  x += 5;
else if (a == 2)
  x += 10;
else if (a == 3)
  x += 16;
else if (a == 4)
  x += 34;
```

Worksheet Four: Repetition Statements

Part One: Counter controlled loops

Question #1 (loops forms):

Write a program to find and print the sum of all numbers between 1 and 1000 using:

- For loop
- While loop
- Do/while loop

Question #2 (Evens and Odds):

Write a program to print the even numbers between 1 and 50 on a massage dialog box, and odd numbers on another one.

Question #3 (Divisibility):

Write a program that reads in two integers and displays all numbers that are odd and divisible by 5. Consider all cases (i.e. num1> num2 or num1<num2)

Question #4 (Series):

Write a java program to calculate and print the value of the following series.

$1+8+27+\ldots\ldots\ldots\ldots + N^3$.

Question #5 (Factorial):

Write a program that reads in an integer, then calculates and prints its factorial.
(Hint: 4! =4*3*2*1 =24)

Question #6 (Prime):

Write a program that reads in an integer and indicates whether this it is Prime or not.

Part Two: Sentinel controlled loops

Question #7 (Average):

Write a program that reads in unspecified number of marks for students, the program should calculate and print the average of all entered marks when the user enters any negative value.

Part Three: Nested loops

Question #8 (Shapes):

Write a program to print the following:

```
*
**
***
****
```

```
 1
22
333
4444
```

Question #9 (Primes) :

Update the program in question 6 to print all the prime numbers from 1 to 100.

Worksheet Five: Working with Methods

Question One (Grades):

1. Write a method **getGrade** that takes the mark as **double** and returns:
    - A: (90<=mark <=100)
    - B: (80<=mark <90)
    - C: (70<=mark <80)
    - D: (60<=mark <70)
    - E: (50<=mark <60)
    - F: otherwise
    - ❖ Hint: use switch statement.

2. Write a method **calculateAverage** that takes the sum of marks as **double**, their number as **integer** and returns the average of marks.

3. Write a method **getResult** that takes the average of marks as **double** and returns "PASS" if the average is greater than or equal 50 and "FAIL" otherwise.

4. Write a method **printResult** that takes the average of marks as **double** and outputs the statement "Congratulations!!! You Passed!!!" if the result is "PASS", and the statement "Sorry!!! You Failed!!!" if the result is "Fail".
    - ❖ Hint: use the **getResult** method with the **equals** or **equalsIgnoreCase** Method.

5. Implement the **main**() method as follows:
    a) Input the unspecified number of marks for a student.
        - ❖ Hint: use a negative value to exit the loop.
    b) Output the grade for each mark.
        - ❖ Hint: call the **getGrade** method.
    c) Output the average of the marks of the student.
        - ❖ Hint: call the **calculateAverage** method.
    d) Output the final result of the student.
        - ❖ Hint: call the **getResult** method.
    e) Output the result statement.
        - ❖ Hint: call the **printResult** method.

Question One (Random):

1. Write a method **calculateRightAnswer** that takes **two integers** and returns their summation.
2. Write a method **checkAnswer** that takes **three integers**. The method should check whether the summation of the first two integers equals the third. If so, the method should return (true), otherwise, the method should return (false).
   - ❖ Hint: use the **calculateRightAnswer** method.
3. Implement the main() method as follows:
   a. Generate **two integer** values **randomly** in the range [1-10].
   b. Input the guessed user answer using Input Dialog:
      - Message: "What is the result of "+ r1 + " + "+ r2+"???!!!! "
      - Title: "Guess??!!!"
      - Icon: QUESTION_MESSAGE
   c. Check whether the sum of the two generated numbers equals the number entered by the user.
      - ❖ Hint: Call the **checkAnswer** method.

   - If so, display a Message Dialog
     - Message: "WOW!!! Right!!!"
     - Title: "Result"
     - Icon: "INFORMATION_MESSAGE"
   - Otherwise, display a Message Dialog
     - Message: "Sorry!!! Wrong"
     - Title: "Result"
     - Icon: "ERROR_MESSAGE"

Worksheet Six: Working with One-dimensional Arrays

Question#1 (Marks)

Write a Java application that uses one dimensional array to do the following:

a) Inputs the number of marks (i.e. array size) from the user
b) Inputs the marks of a student into the array
c) Prints the marks of the student
d) Calculates and displays the average of marks
e) Finds out the number of marks that are above the average
f) Write a method that finds and returns the minimum mark
g) Write a method that finds and returns the maximum mark
h) In the main(), display the minimum and the maximum mark
i) Rewrite the program using dialog boxes.

Question#2 (Even_Odd)

Write a java program that :

a) Define a method called **sumOfGreaters (int [], int)** that finds the sum of all elements that is greater than second parameter.

b) Define a method called **FillEven(int[],char [])** that checks whether each number in the integer array is even or odd and fills the corresponding position of the character with either 'E' or 'O'.

Example:

If the user entered 15 - 24 the number array is:

15	16	17	18	19	20	21	22	23	24

The Even array must be as:

O	**E**	**O**	**E**	**O**	**E**	**O**	**E**	**O**	**E**

c) In the main():
   a. Prompt the user to input two integer's **num1 and num2** and define an integer array called **"numbers"** that contains numbers from num1 to num2.
   b. Prompt the user to input an integer and output the sum of all array elements that are greater that integer.
   c. Create a character array and fill it with either 'E' or 'O' using the already defined method.

Worksheet Seven: Working with Two-dimensional Arrays

Question#1 (Marks)

a) In the main(), create and initialize the following two-dimensional array, name it "array".

1	2	3	4
7	8	2	2
9	8	7	6
1	0	1	2

b) Define the method named getSumOfArray(int [][] a) that calculates and returns the sum of all elements of the array.

c) Define the method named getSumofRow3 (int [][] a) that calculates and returns the sum of the third row of the array.

d) Define the method named getSumofCol3 (int [][] a) that calculates and returns the sum of the third column of the array.

e) Define the method named getSumofDiagonal (int [][] a) that calculates and returns the sum of the primary diagonal of the array.

f) Define the method named printRowsSum (int [][] a) that calculates and prints the sum of each row of the array.

g) Define the method named printColsSum (int [][] a) that calculates and prints the sum of each column of the array.

Question Two (ASCII):

a) Create an integer two-dimensional array named "ascii"; the dimensions of this array should be entered by the user.

b) Define the method fillAsciiArray (int [][]ascii) that fills the ascii array with random integer values between 65 and 100.

c) Define the method fillLettersArray (int [][]ascii) that creates a character array "letters" of the same size of the integer array, and fills it with the letter that corresponds to the ascii code in the corresponding position in the ascii array. The method should return the letters array.

d) In the main():
   a. Print out ascii array in a tabular format. (Hint: the dimensions should be entered by the user).
   b. Print out the letters array in a tabular format.

Worksheet Eight: Working with Classes

Question#1 (Constructors):

1. Create a class called **Rectangle** that contains the following data fields:
    - **height** as integer.
    - **width** as integer.
2. The class also contains the following methods:
    - A no-arguments **constructor** that initializes **Rectangle** data fields to their default values.
    - A two-arguments **constructor** that initializes **Rectangle** data fields to specific given values.
    - A method named **getArea()** that returns the area of the rectangle.

3. In the **main** method:
    - Create an object (r1) with its default values
    - Create an object (r2) of the **Rectangle** class with width=10 and height =20.
    - Print the area of the rectangle r1 by invoking getArea().

Question#2 (Access Modifies, Mutators and Accessors):

1. Create a class called **Student** that contains the following data fields:
    - **name** as private String.
    - **stdno** as private integer.
    - **grade** as private integer.
2. The class also contains the following methods:
    - A no-argument c**onstructor** to initialize the **Student** data fields to its default values.
    - Set and get methods for all data fields.
        - **setGrade** method will check if the grade is less than 40 it will be set to 40, otherwise the grade will be the same.
    - **classifyGrade** method that has no argument and returns the appropriate **letter** for the grade according to the following:
      80 - 100 --------- A
      50 – 79 ---------- B
      < 50 ---------- F
    - **print** method that prints the **Student** data fields and the grade letter (invoke classifyGrade method).
3. In the **main** method :
    - Create an object (s1) of the **Student** class with name ="Ahmad", ID= 123415, grade=85
    - Invoke **classifyGrade** method and print the corresponding grade letter for s1.
    - Print all student information.

Question#3 (Instance variables and static variables):

1. Create a class called **Account** that contains the following data fields:
    - **num** as a static private integer
    - **id** as private integer.
    - **balance** as private double.
2. The class also contains the following methods:
    - A **no-arguments constructor** that
        - initializes data fields to their default values
        - increments **num** and assigns the new **num** to the **id**.
    - The **set** and **get** methods for **id** and **balance** data fields.
    - **Static set** and **get** method for **num** data field.
    - A method called **withdraw(double)** that withdraws a specified amount from the account **balance**. Before withdraw, the method should check whether the available balance is sufficient.
    - A method named **deposit(double)** that deposits a specified amount to the account **balance**.
3. In the **main** method :
    - Create an object (a1) of the **Account** class with balance=1000.
    - Create an object (a2) of the **Account** class with balance=2000.
    - Print id and balance of a1 and a2.
    - Withdraw 100 from a1 balance.
    - Deposit 50 to a1 balance.
    - Print the balance of a1 and a2.
    - Print the total number of the created accounts

Worksheet Nine: Working with String Methods

Basic String Methods API

- int length(): It returns the length of a String.
- String toUpperCase(): Converts the string to upper case string using the rules defined by specified locale.
- boolean equals(Object obj): Compares the string with the specified string and returns true if both matches else false.
- boolean equalsIgnoreCase(String string): It works same as equals method but it doesn't consider the case while comparing strings. It does a case insensitive comparison.
- char charAt(int index): It returns the character at the specified index. Specified index value should be between 0 to length() -1 both inclusive. It throws IndexOutOfBoundsException if index<0||>= length of String.
- int indexOf(int ch): Same as indexOf method however it starts searching in the string from the specified fromIndex.
- String substring(int beginIndex): It returns the substring of the string. The substring starts with the character at the specified index.
- String substring(int beginIndex, int endIndex): Returns the substring. The substring starts with character at beginIndex and ends with the character at endIndex.
- String replace(char oldChar, char newChar): It returns the new updated string after changing all the occurrences of oldChar with the newChar.
- boolean contains(CharSequence s): It checks whether the string contains the specified sequence of char values. If yes then it returns true else false. It throws NullPointerException of 's' is null.
- public boolean isEmpty(): This method returns true if the given string has 0 length. If the length of the specified Java String is non-zero then it returns false.
- String[] split(String regex): It splits the string and returns the array of substrings that matches the given regular expression.

Question#1 (using String Methods):

Create a Java application **(Q1)**.

- The application should repeatedly ask the user to enter a word (using input box) until the user enters the word "stop". When the user enters "stop":
- The number of words that contains the letter "m" will be displayed on a message box.
- The number of letters in the shortest word will be displayed on a message box.

Question#2 (Array of Strings):

Create a Java application **(Q2)**.

- The application should ask the user to enter sequence of nouns in the console separated by either commas or spaces. The application should read the sequence of nouns as line.
- Create a String array (nouns) that contains the nouns in the line.
- Create a Boolean array (isVowel) with the same length as nouns.
- Create a String array (articles) with the same length as nouns
- Fill the array isVowel with true if the first character of each noun in the corresponding position in the array nouns is a Vowel (capital or small), otherwise fill that position with false.
- Fill the array articles with "An" if the first character of each noun in the corresponding position in the array nouns is a Vowel, otherwise fill that position with "A".
- Print the elements in the three arrays in a tabular format.

```
run:
Enter sequence of nouns separated by either commas or spaces
book orange car,Apple Home
book    false   A
orange  true    An
car     false   A
Apple   true    An
Home    false   A
BUILD SUCCESSFUL (total time: 41 seconds)
```

Worksheet Ten: Working with Files

Question#1 "Writing to Files"

Write a Java application that writes your name, your ID, and your major to a file named "myfile.txt"

Question#2 "Reading from Files"

Write a Java application that read the content of a text file line by line and display it on the Console.

Question#3 "Copy File Content"

Write a Java application that copy the content of a text file into another text file.

Question#4 "Comma-separated file"

A student text file has the following structure "FirstName,MiddleName,LastName,Mark".

Write a Java code that:

a) Writes the firstName and the lastName of each students in a new text file.
b) Counts the number of students and displays it on the console.
c) Finds the maximum mark and displays it on the console.